



Building AI Scientist Workflows for Molecular Discovery

with BioNeMo Agent Toolkit

Kyle Tretina, NVIDIA · ACS Fall 2026 Workshop



Getting started

<https://bit.ly/4xS21BK>



Preview Code Blame 161 lines (120 loc) · 7.93 KB

  Raw    

ACS Fall 2026 GPU chemistry workshop

Explore a bounded chemistry-agent workflow with the NVIDIA nvMolKit library. You can use an interactive notebook, a conversational OpenClaw experience, or both. Each journey uses fixed inputs and validated operations so that you can focus on the chemistry and the scientific limits of the results.

Before you begin

- Use a current desktop browser.
- Brev GPU time is billable. Stop your environment when you finish.
- Keep API keys private. Enter a key only in the named Launchable field. Never paste it into a notebook, chat, or workshop result.
- Hosted inference can be rate-limited. If a request reports an error, take a screenshot and continue with another optional activity.

Chemical and Materials Industry Is a Multi-trillion Dollar Market

Global market size in 2022, USD billion — yet a new material still takes 10-20 years to reach market



Pharmaceuticals & BioPharma

Small-molecule medicines
Biologics (e.g., antibodies, recombinant proteins)
Vaccines
Cell and gene therapies



Basic chemicals

Organic (e.g. ethylene, methanol)
Inorganic (e.g. hydrogen, chlorine)



Specialty chemicals

Electronic materials (e.g. semiconductor, battery, OLED)
Paints and coatings
Flavors and fragrances
Catalysts



Plastics and fibers

Commodity plastics (e.g. PE)
Fibers (e.g. polyester)
Engineering plastics (e.g. ABS)
High-performance polymers (e.g. PTFE)



Agrochemicals

Fertilizer (e.g. ammonia)
Pesticide

The Discovery Bottleneck

Chemical space is effectively infinite — synthesis, validation, and compute are not

SEARCH SPACE

$\sim 10^{60}$

Plausible drug-like molecules, plus an equally vast inorganic composition space. No experimental program can sample a meaningful fraction of it

SCREENING CEILING

$10^3 - 10^4$

Candidates per campaign under conventional DFT workflows. The funnel narrows long before the chemistry does

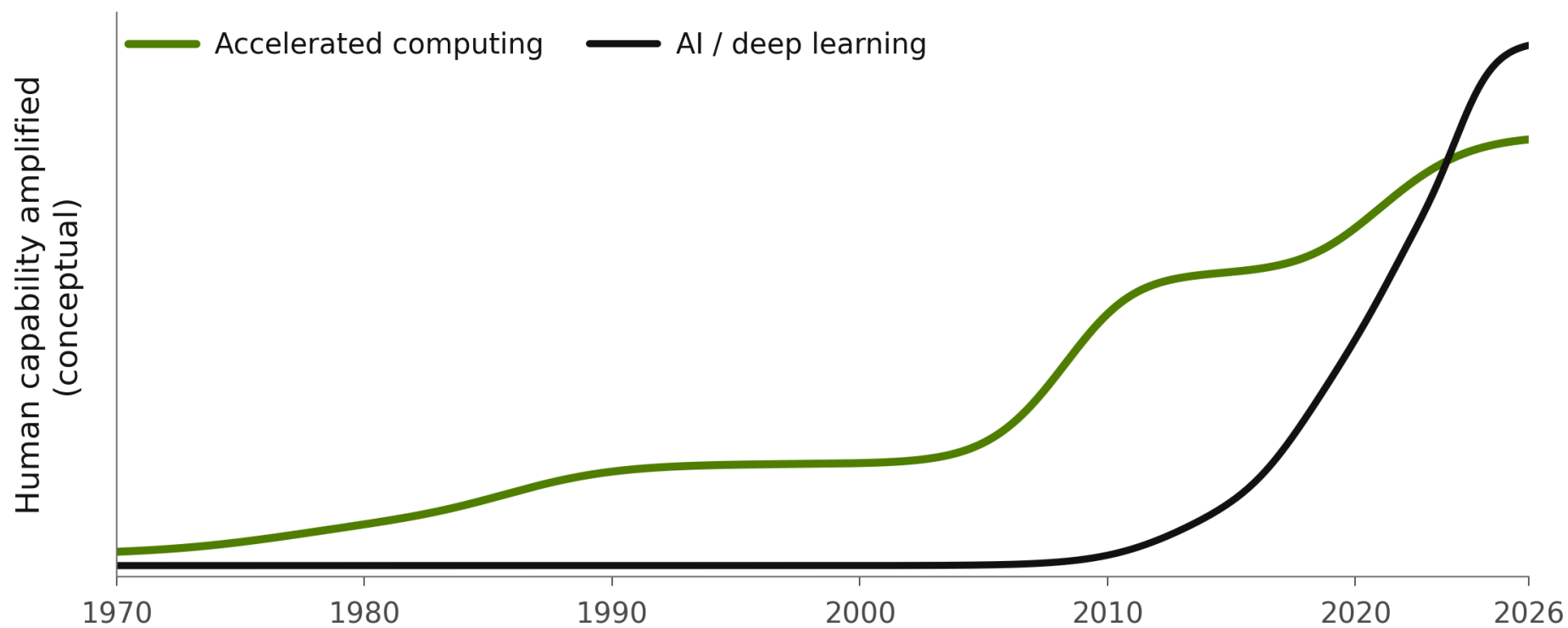
TIME TO MARKET

10-20 yrs

From concept to a commercial material, paced by synthesize-test-analyze loops measured in weeks each

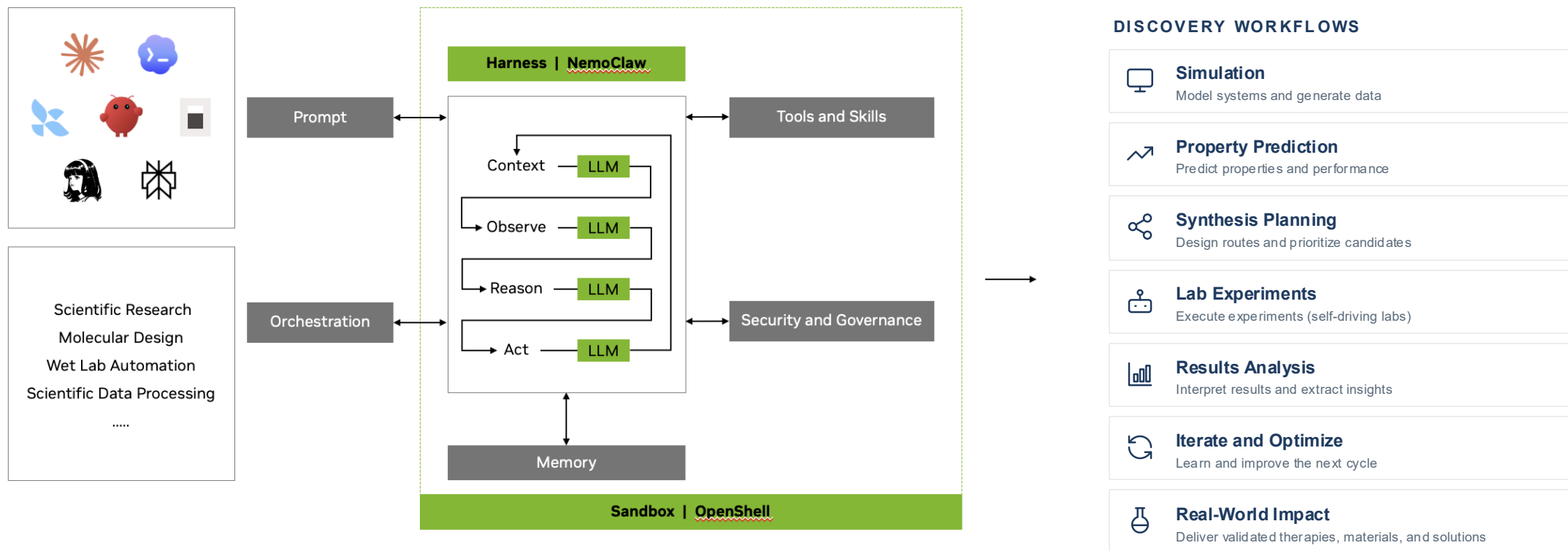
Two Revolutions Are Accelerating Chemistry

Accelerated computing expands what chemists can simulate; AI expands what they can predict and design



Can AI act as a scientific researcher, not just assist one?

For today, not without the right tools



NVIDIA Enables the Chemistry Agent Stack

An open, modular stack for building scientific agents and chemistry discovery workflows

MODELS AND INFERENCE

LLMs

Reasoning and multimodal models for planning, interpretation, and scientific interaction.

NeMo Switchyard

Task-aware model routing across endpoints to optimize quality, latency, and cost.

AGENT TOOLS & SKILLS

BioNeMo Agent Toolkit

Scientific orchestration for multi-step life sciences and chemistry tasks.

Related Skills

Scientific tools, structured databases, retrieval, and domain-specific actions.

Guardrailing

Policy, permission, safety, and execution controls across the agent loop.

NeMo Relay

Observability for agent runs, tool calls, model calls, and scientific traces.

Skill Evaluator

Measures task quality, scientific validity, safety, and reproducibility.

NeMo Gym + RL

Scalable environments and feedback-driven improvement for agent behavior.

REFERENCE HARNESS & BLUEPRINTS

NeMoClaw

Base harness for custom scientific agents, domain context, and tool orchestration.

AI-Q / BioNeMo Deep Research

Deep research and enterprise knowledge workflows for scientific discovery.

SECURE RUNTIME

OpenShell

Policy-based controls for networks, tools, files, code execution, privacy, and secrets.

What Is BioNeMo Agent Toolkit?

A toolkit for building agentic AI solutions across the life sciences value chain

Agent

 **AGENT** Plans · Reasons · Orchestrates Work

Skills

Skills

Reusable capabilities

DIGITAL BIOLOGY

- NIM microservice skills
- Library skills
- Open model skills
- Workflow skills

Blueprints

Reference workflows

DIGITAL BIOLOGY

- Generative Protein Binder Design
- Generative Virtual Screening
- Evo 2 Protein Design

Inference

Model endpoints

NVIDIA NIM MICROSERVICES

- OpenFold2, OpenFold3 · Boltz-2
- DiffDock · GenMol · MolMIM
- MSA Search · Evo 2
- RFdiffusion · ProteinMPNN
- Nemotron Reasoning

Tools

Enabling agent building

- NeMo
- NeMoClaw
- OpenShell

Tools

Open protein & molecule
datasets · Genomic &
transcriptomic corpora ·
Synthetic structures (CDSPD)



Data

nvDock · ReaSyn · AvgFlow ·
CodonFM · RNAPro · Proteina ·
La-Proteina · GenMol ·
ProtComposer · Megalodon ·
DualBind · KERMT



Models

nvMolKit · cuEquivariance ·
cuik-molmaker ·
nvSubquadratic · WARPkan ·
nvQSP · scdl · Parabricks · cuDF
· cuML · cuGraph · cuVS ·
nvCOMP · cuEST



Libraries

Fine-tuning workflows · Model
adaptation · LoRa · low
precision · parallelism



Recipes



nvMolKit

GPU-accelerated chemistry tools

Kevin Boyd, NVIDIA · ACS Fall 2026 Workshop

nvMolKit – GPU extensions to the RDKit Cheminformatics Package

Why target RDKit?

RDKit is the industry standard for cheminformatics, but many of its functions are bottlenecks in high-throughput machine learning workloads. Targeting RDKit accelerates a widely used, well-understood framework and unlocks faster performance in large-scale AI and scientific applications.

Fully Open-Source and Buildable

- Open source
 - following RDKit's ethos to serve a massive community of developers and scientists, with public contributions improving the library collaboratively
- Buildable
 - advanced users tailor the code to their hardware and needs, with distributed packages on PyPI and conda

nvMolKit Architecture

Core C++/CUDA library

A standalone C++/CUDA library with Python bindings, linking directly to the RDKit C++ API for full functional access with minimal overhead

Optimized for batches

A near drop-in replacement for RDKit APIs, with one key difference: it is built to handle batches of operations for significant speedup

Seamless integration

The same Boost.Python bindings as RDKit let RDKit data structures pass directly into nvMolKit functions

Open source

Free, fully open source including the CUDA code, permissively licensed, and buildable directly from source

Upstreamed RDKit Accelerations

Goal: accelerate RDKit functions that are common bottlenecks in ML workloads — all CPU work upstreamed to RDKit

2.4x

Morgan fingerprints — up to 2.4x on common workloads, scaling with radius

30%

SMILES parsing — linear speedup, up to 30% for a 200-atom molecule

12x

Butina clustering — up to 12x on common workloads

4–8%

MMFF minimization — 4-8% improvement in serial

5%

Conformer generation — 5% improvement in serial

Ongoing SMILES Parsing + Sanitization improvements - github.com/rdkit/rdkit/issues/9475

nvMolKit: GPU-Accelerated RDKit

Goal: accelerate RDKit functions that are common bottlenecks in ML workloads

7x

Morgan fingerprinting, single GPU

3000x

Tanimoto similarity, many-to-many cross

1750x

Cosine similarity, many-to-many cross

1000x

Butina clustering on Tanimoto distances

35x

MMFF minimization at ~100 atoms, scaling to larger molecules

16x

Conformer generation at ~100 atoms, scaling to larger molecules

30x

Substructure search across core operations

Substructure operations: countSubstructMatches · hasSubstructMatch (validated with the BMS 2006 filter) · getSubstructMatches (validated with the tautomer enumerator)

Tanimoto and Cosine Similarity

DESCRIPTION

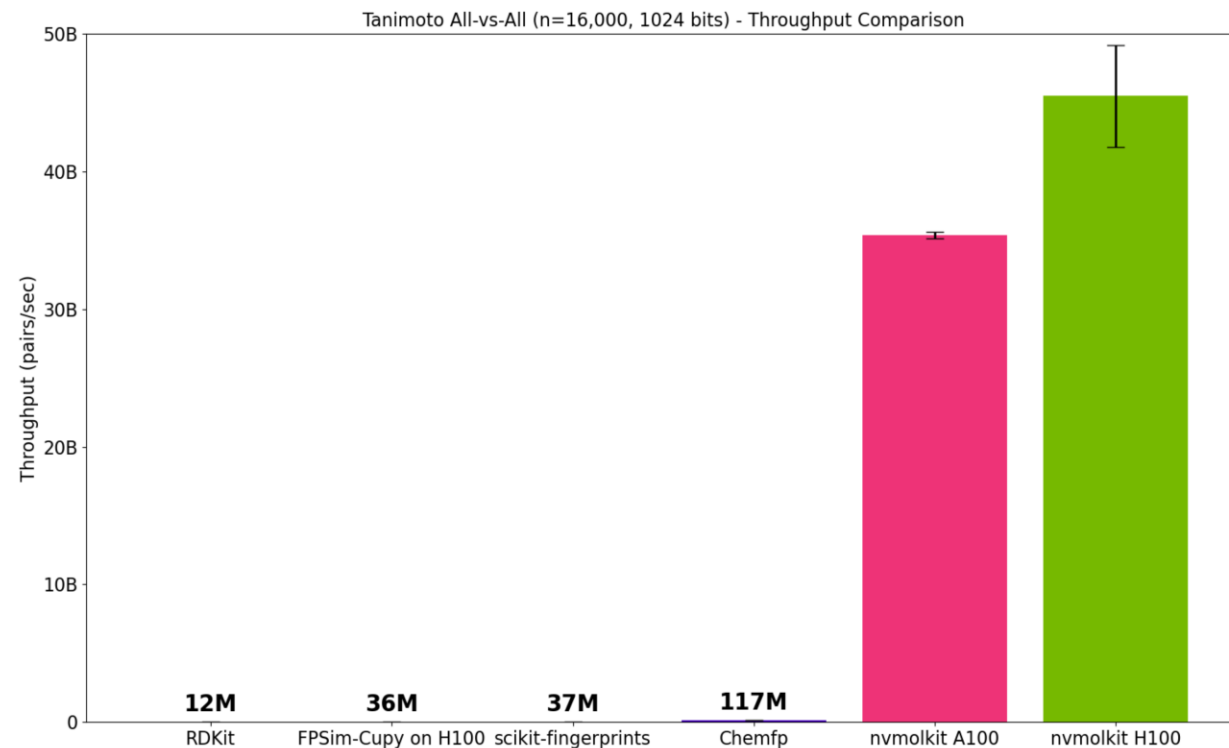
- Tanimoto similarity is a popular metric for the distance between two molecules based on binary fingerprints
- Cosine similarity is more general, comparing molecules as numerical vectors such as count-based fingerprints or embeddings

SIGNIFICANCE

Tanimoto has been used extensively as a MolMIM oracle, and both metrics search large chemical databases for similar molecules

OUR WORK

- Up to 3000x vs RDKit on cross-Tanimoto and 1750x on cross-cosine when input data is already on the GPU
- >300x faster than state of the art non-RDKit implementations, including GPUs



MMFF Geometry Relaxation

DESCRIPTION

The Merck Molecular Force Field is a molecular mechanics force field that optimizes 3D geometry by minimizing potential energy — relaxing strain, removing steric clashes, and refining bond lengths and angles into a physically plausible conformation

SIGNIFICANCE

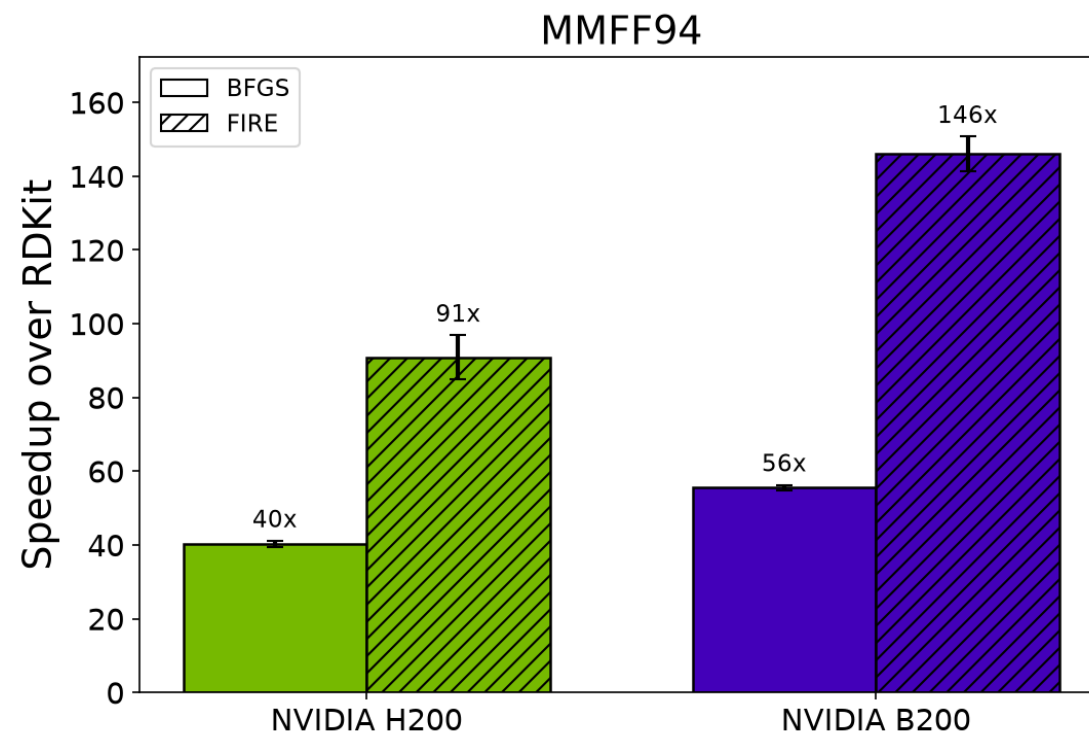
A critical post-processing step for conformer generation — it keeps the ensemble chemically realistic, ranks conformers by relative stability, and filters out high-energy structures

OUR WORK

56x speedup on drug-like molecules (Enamine REAL) vs multithreaded CPU, using BFGS minimizer

146x speedup using FIRE minimizer

MMFF94 force-field optimization speedup on 8 GPUs over RDKit (112-thread Intel Xeon Platinum 8480C)



ETKDG Conformer Generation

DESCRIPTION

ETKDG is a geometry-based algorithm that generates a diverse ensemble of plausible 3D conformers by embedding a molecule into three-dimensional space using a distance matrix derived from bond, angle, and torsion data

SIGNIFICANCE

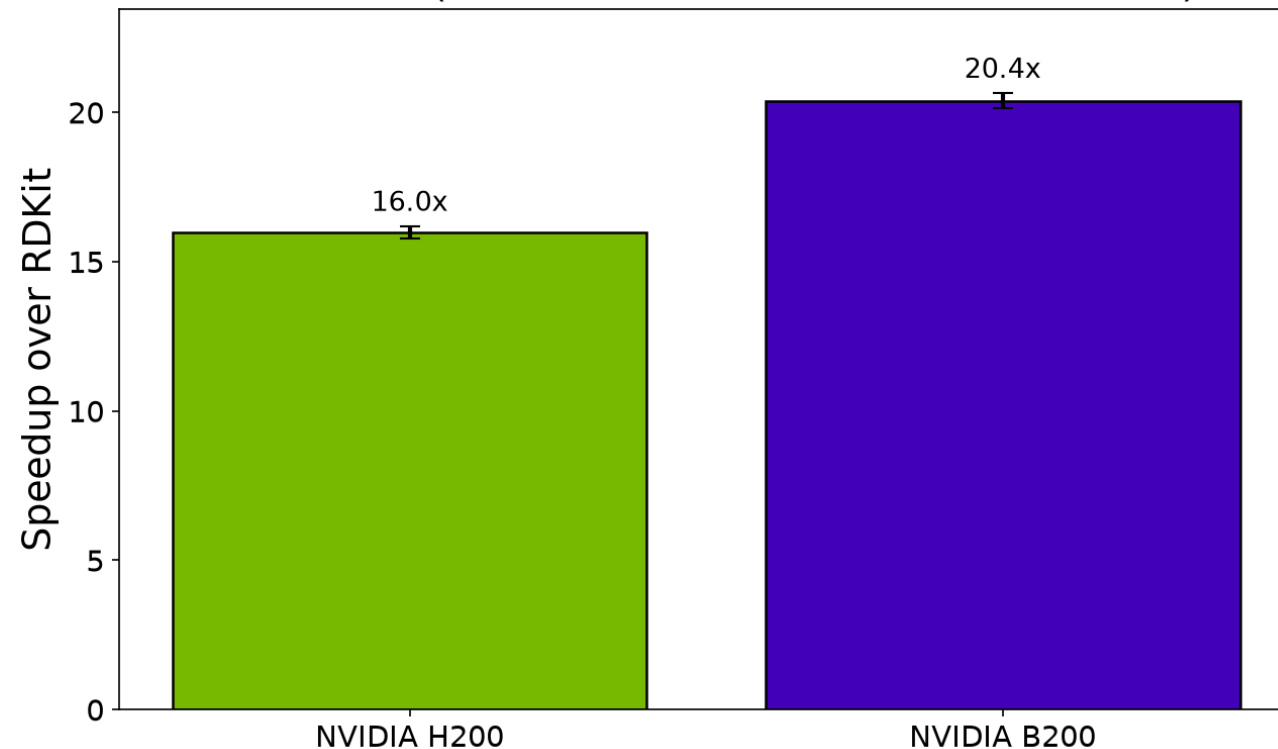
A prerequisite for any computational chemistry task that needs 3D structures — molecular docking, 3D pharmacophore modeling, and preparing molecules for 3D-aware machine learning

OUR WORK

20x

speedup on drug-like molecules (e.g. ChEMBL);
scales linearly with molecule type

ETKDG speedup on 8 GPUs
over RDKit (112-thread Intel Xeon Platinum 8480C)



Butina Clustering

DESCRIPTION

Taylor-Butina clustering is an iterative unsupervised algorithm for clustering molecules based on pairwise distances (typically Tanimoto or similar). It iteratively selects the largest possible cluster within a distance threshold.

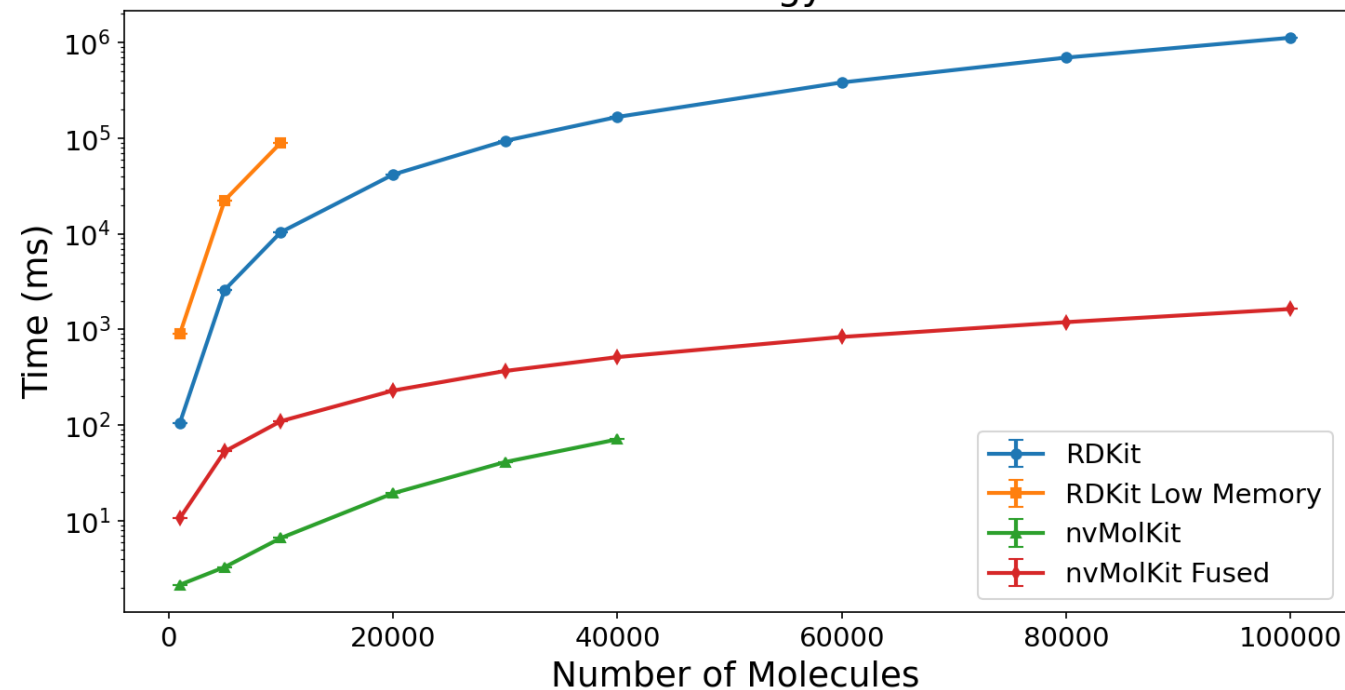
SIGNIFICANCE

Clustering is important for characterizing molecular datasets, as well as for selecting splits for ML datasets

OUR WORK

>1000x Speedup on large datasets on an H200 vs single CPU

Butina Clustering Time on H200 (Cutoff 0.2)
Semilogy



Substructure Search

DESCRIPTION

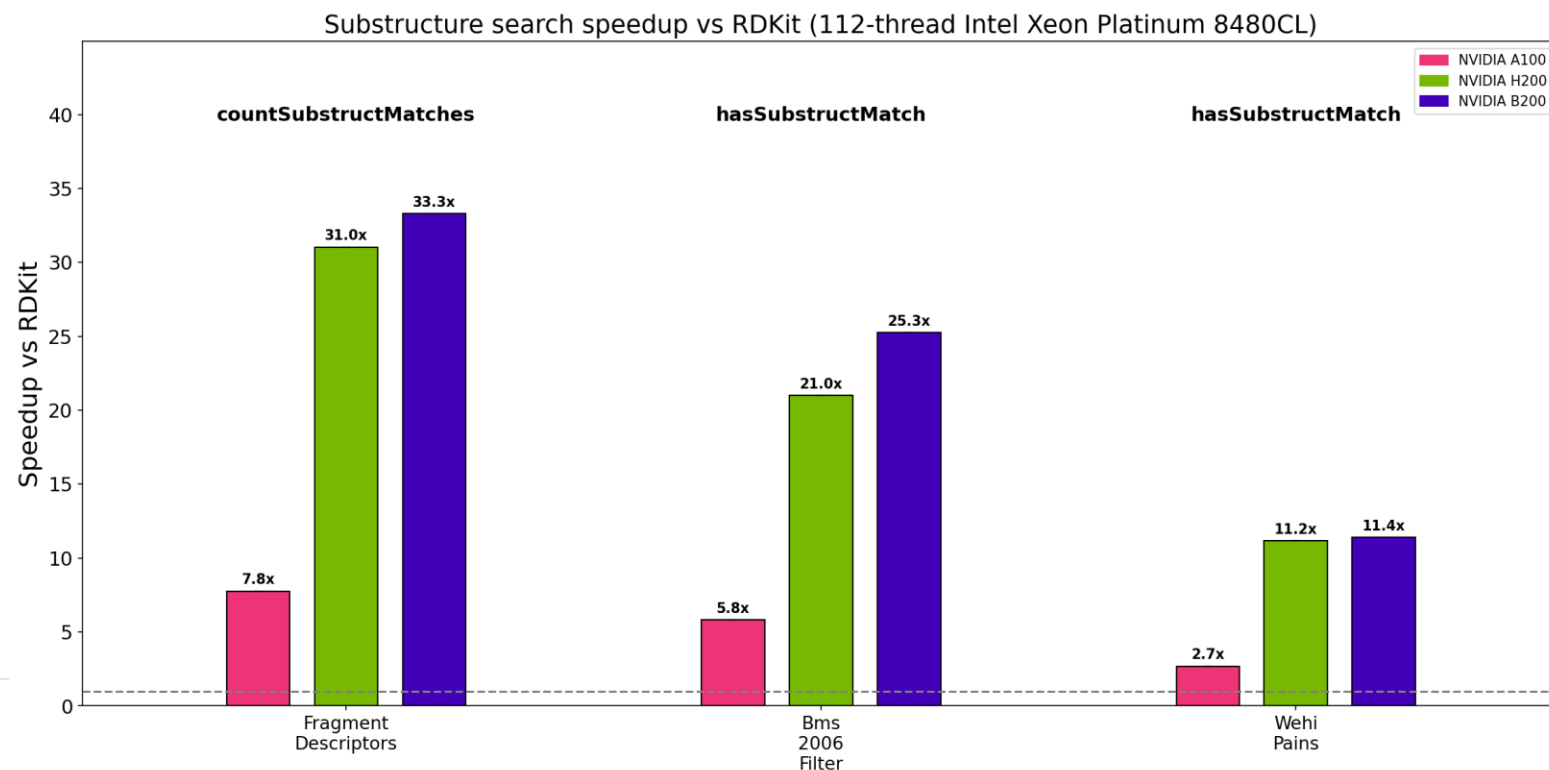
Substructure search is a fundamental cheminformatics task that determines whether a specific query pattern exists within a larger target molecule.

SIGNIFICANCE

Substructure searching is essential for chemical library filtering, identifying specific pharmacophores or functional groups, and analyzing chemical series in lead optimization. High-performance batch searching is a critical prerequisite for screening massive molecular databases to identify compounds that contain specific chemical motifs or structural requirements.

OUR WORK

11-33x Speedups on B200 vs multithreaded RDKit



Maximum Common Substructure

DESCRIPTION

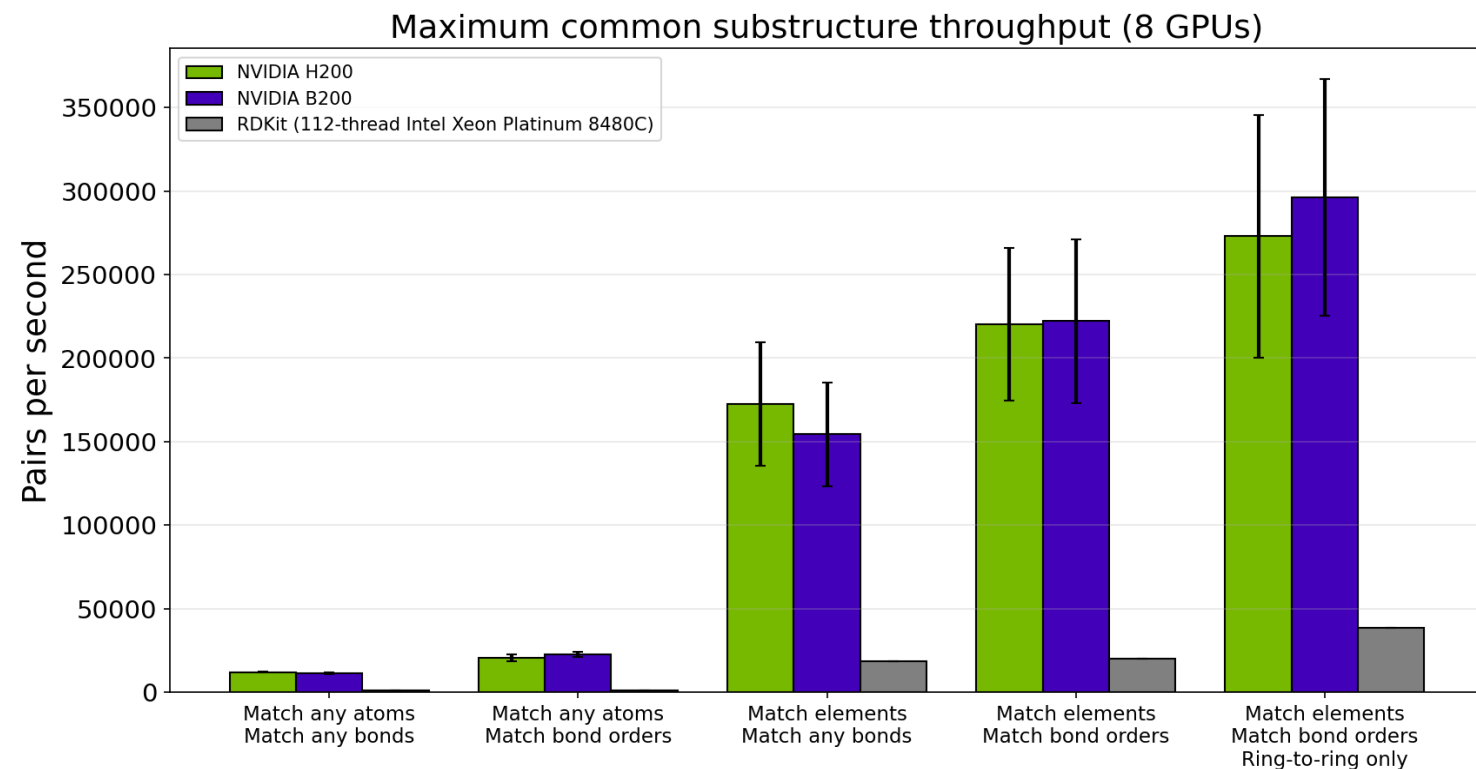
Finds the largest common subgraph between two molecules. A very expensive operation compared to general substructure search.

SIGNIFICANCE

Used for high fidelity similarity calculations, clustering, molecular alignment, molecular scaffolding, and reaction mapping.

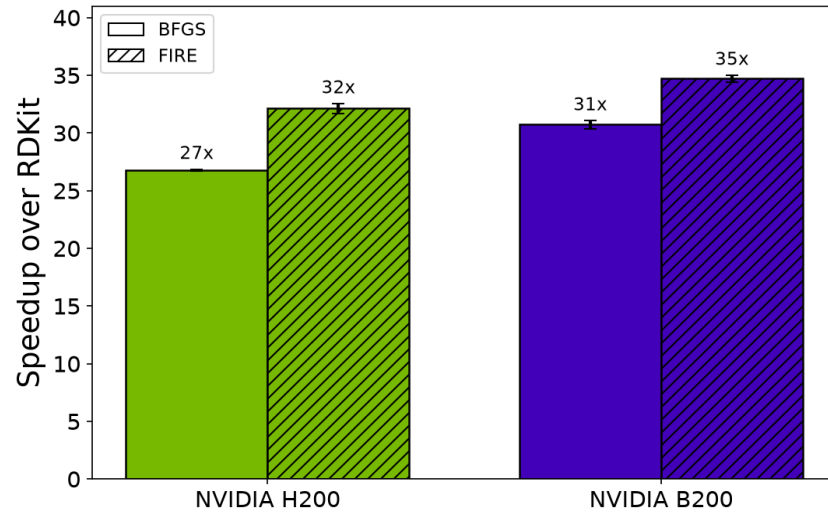
OUR WORK

10-17x Speedups on B200s vs multithreaded RDKit

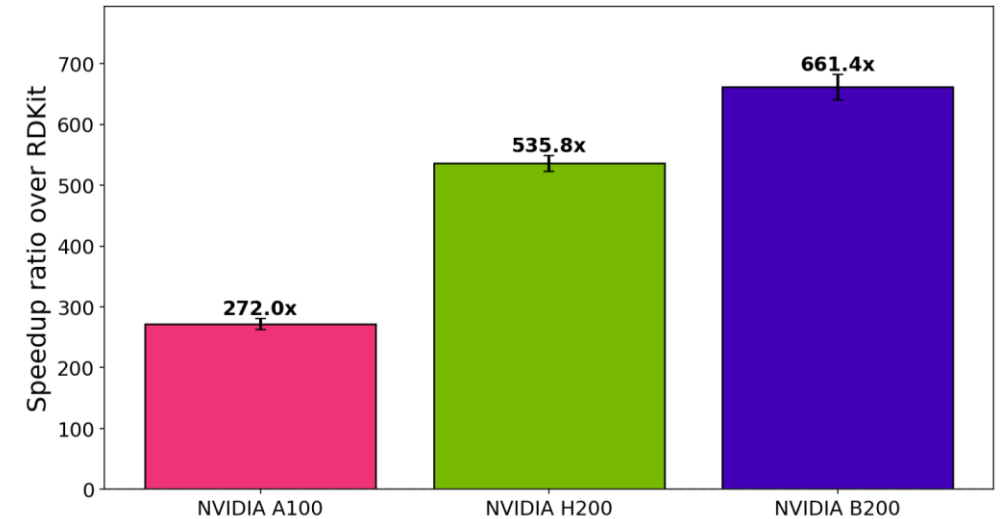


Other Accelerations

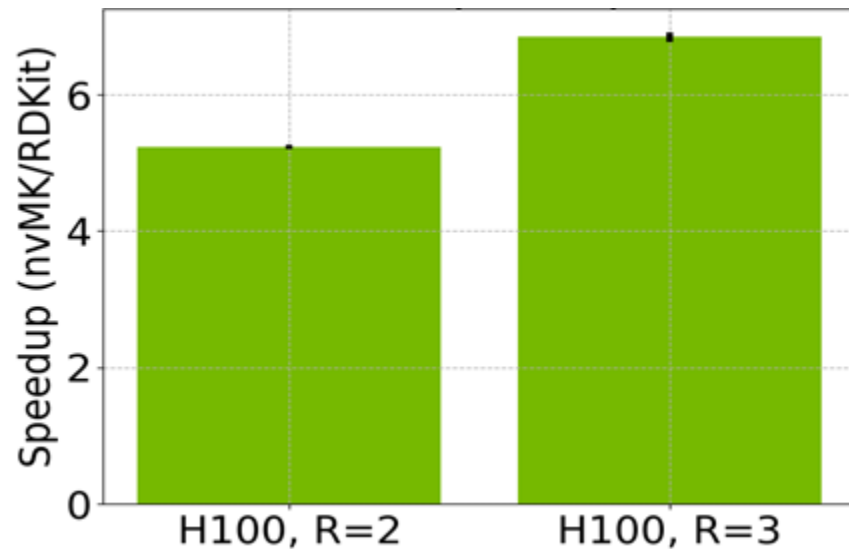
UFF Optimization (35x)



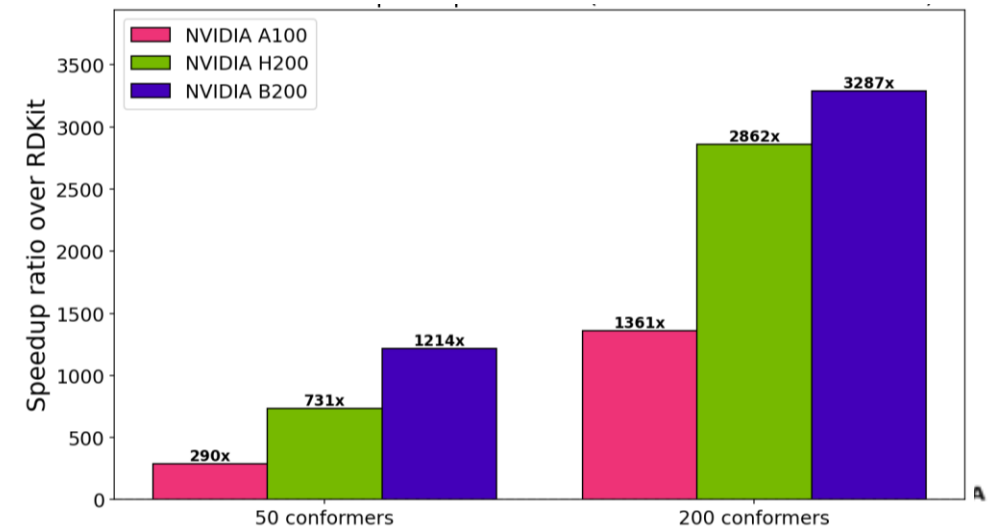
Torsion Fingerprint Deviation (600x)



Morgan Fingerprints (6x)



Conformer RMSD Matrix (3000x)



nvMolKit Resources

Repository

github.com/NVIDIA-Digital-Bio/nvMolKit

PyPI

pypi.org/project/nvMolKit

Blog post for v0.1

research.nvidia.com/labs/dbr/blog/nvMolKit

Documentation

nvidia-digital-bio.github.io/nvMolKit

conda-forge

anaconda.org/conda-forge/nvmolkit

Releases

[nvMolKit release tracker](#)



Demo 1: Notebooks

Ian Kerman, NVIDIA · ACS Fall 2026 Workshop



Demo 2: Interactive



GET STARTED TODAY

Build on the BioNeMo Platform

Tools and libraries, open models and data, NIM microservices

NVIDIA BioNeMo Agent Toolkit

github.com/NVIDIA-BioNeMo/bionemo-agent-toolkit

NVIDIA-BioNeMo on GitHub

github.com/NVIDIA-BioNeMo

